

Graph Theory – Classifying Graphs

Name _____ Per _____

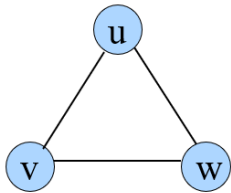
Graph: collection of vertices and edges

Undirected graph: all edges are assumed to be bidirectional

WAYS TO REPRESENT AN UNDIRECTED GRAPH: NODE & EDGE LISTS, ADJACENCY LIST, AND ADJACENCY MATRIX

Node list: {u,v,w}

Edge list: {(u,v), (u,w), (v,w)}



node	Adjacency List
u	v , w
v	w, u
w	u , v

	v	u	w
v	0	1	1
u	1	0	1
w	1	1	0

Adjacency matrix: matrix containing 1's and 0's where a 1 indicates an edge exists between the vertices to the left of the that cell and above the cell. So the first 1 in the first row indicates v is connected to u.

The adjacency matrix for an undirected graph will always be symmetric about the major diagonal.

Degree: total number of connections to a vertex, whether inbound or outbound

Isolated vertex: a vertex with degree zero; that is, a vertex that is not an endpoint of any edge

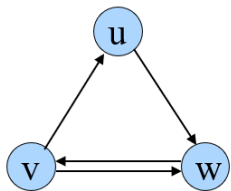
Leaf vertex: a vertex with degree one.

Directed graph: direction of edges are indicated

WAYS TO REPRESENT AN DIRECTED GRAPH: NODE & EDGE LISTS, ADJACENCY LIST, AND ADJACENCY MATRIX

Node list: {u,v,w}

Edge list: {(u,w), (v,u), (v,w), (w,v)}



node	Adjacency List
u	w
v	u, w
w	v

	v	u	w
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Indegree: number of inbound connections

Outdegree: number of outbounds connections

Source vertex: a vertex in a directed graph with indegree zero

Sink vertex: a vertex a directed graph with outdegree zero

Simple Graph: doesn't contain multiple edges between nodes nor loops. A **simple directed graph** has no multiple arrows with same source and target nodes.

Non-simple Graph: contains multiple edges between at least one pair of nodes OR has at least one loop.

Connected graph: path exists from every vertex to every other vertex

Unconnected graph: a graph that is not connected and is made up of connected components

Walk: list of vertices in which successive vertices are connected by edges: SB-MK-F1

Closed Walk: a walk that starts and ends at the same vertex
Sponge Bob-Patrick-Karen-Patrick-Sponge Bob

Trail: a walk which has no repeating edges:
Gary-Patrick-SpongeBob-Gary-Karen-Plankton

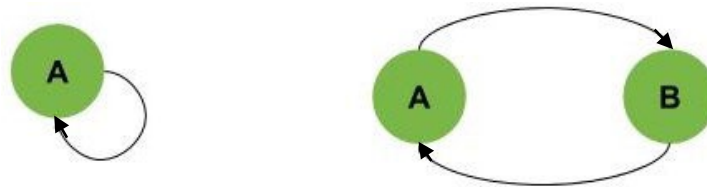
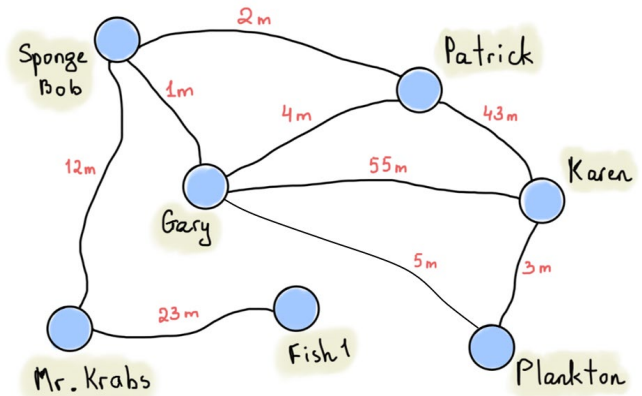
Circuit: a closed trail (one which ends where it started)
SBob-Patrick-Gary-Karen-Plankton-Gary-SBob

Path: a trail which has no repeating vertices, except may the first and last:
SpongeBob-Gary-Karen

Cycle: a closed path (one which ends where it started)
Gary-Patrick-Karen-Gary

Directed cycle: The same definition as above, but since each directed edge is distinct, you could have a cycle of length two, whereas in an undirected graph you can have cycles of length one (if there's a loop) or length three, but not length two as that would necessitate traversing the same edge twice.

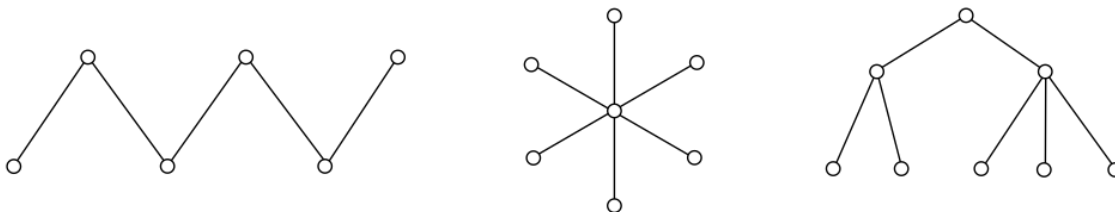
A simple undirected weighted cyclic graph:



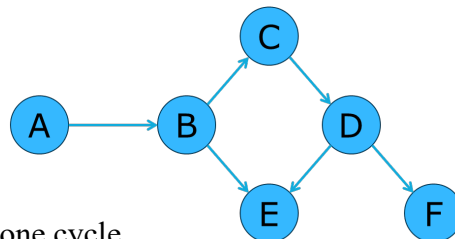
Remember that in directed graphs, each arrow is considered to be a separate connection. So AB and BA are distinct edges. This means ABA would count as a cycle!

Tree: An undirected graph in which any two vertices are connected by exactly one path.

I.e. a connected acyclic undirected graph. I.e. a connected graph that does not contain even a single cycle.



Directed Acyclic Graph (DAG):
a directed graph with no cycles



Cyclic Graph: A graph with at least one cycle.

Circular Graph: A graph that is a cycle.

Complete Graph: A simple graph in which each pair of distinct vertices are adjacent.

